

Hamming Weights of Integers

The Hamming weight of a number is the **number of set bits** (1-bits) in its binary representation. Given a positive integer n , return an array where the i th element is the Hamming weight of integer i for all integers from 0 to n .

Example:

Input: $n = 7$
Output: $[0, 1, 1, 2, 1, 2, 2, 3]$

Explanation:

Number	Binary representation	Number of set bits
0	0	0
1	1	1
2	10	1
3	11	2
4	100	1
5	101	2

6	110	2
7	111	3

Intuition - Count Bits For Each Number

The most straightforward strategy is to individually count the number of bits for each number from 0 to n.

Consider a number $x = 25$ and its binary representation:

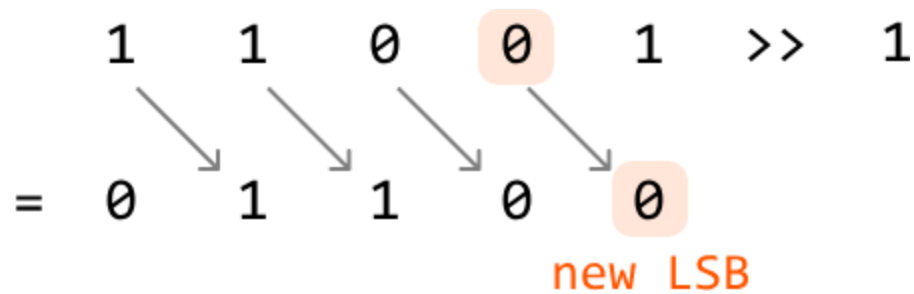
$$\begin{aligned}
 x &= 25 \quad (\text{base } 10) \\
 &= 1 \quad 1 \quad 0 \quad 0 \quad 1 \quad (\text{base } 2)
 \end{aligned}$$

To count the number of set bits (1s) in a number, we can check each bit and increase a count whenever we find a set bit. Let's see how this works.

For starters, we can determine the least significant bit (LSB) of x by performing $x \& 1$, which masks all bits of x except the LSB: if $x \& 1 == 1$, the LSB is 1. Otherwise, it's 0. We can see this below for $x = 25$:

$$\begin{array}{r}
 \begin{array}{cccccc}
 1 & 1 & 0 & 0 & 1 & (x) \\
 \& 0 & 0 & 0 & 0 & 1 & (1) \\
 \hline
 0 & 0 & 0 & 0 & 1 & (1)
 \end{array}
 \end{array}$$

Now, how do we check the next bit? If we perform a bitwise right-shift operation on x , we shift all bits of x one position to the right. This effectively makes this next bit the new LSB:



We now have a process we can repeat to count the number of set bits in a number:

1. If $x \& 1 == 1$, increment our count. Otherwise, don't.
2. Right shift x .

Continue with the two steps above until x equals 0, indicating there are no more set bits to count.

Doing this for every number from 0 to n provides the answer.

Try it yourself

Write your solution to the problem before checking the reference implementation.

Implementation - Count Bits For Each

Number

Java:

```
import java.util.ArrayList;
```

```
public class Main {  
    public ArrayList<Integer> hamming_weights_of_integers(int n) {  
        ArrayList<Integer> result = new ArrayList<>();  
        for (int x = 0; x <= n; x++) {  
            result.add(count_set_bits(x));  
        }  
        return result;  
    }  
}
```

```

}

public int count_set_bits(int x) {
    int count = 0;
    // Count each set bit of 'x' until 'x' equals 0.
    while (x > 0) {
        // Increment the count if the LSB is 1.
        count += x & 1;
        // Right shift 'x' to shift the next bit to the LSB position.
        x >>= 1;
    }
    return count;
}
}

```

Complexity Analysis

Time complexity: The time complexity of `hamming_weights_of_integers` is $O(n \log(n))$ because for each integer x from 0 to n , counting the number of set bits takes logarithmic time, as there are approximately $\log_2(x)$ bits in that number. If we assume all integers have 32 bits, the time complexity simplifies to just $O(n)$, since counting the set bits for a number will take at most 32 steps, which we do for $n+1$ numbers.

Space complexity: The space complexity is $O(1)$ because no extra space is used except the space occupied by the output.

Intuition - Dynamic Programming

In the previous approach, it's important to note that by the time we reach integer x , we have already computed the result for all integers from 0 to $x - 1$. If we find a way to leverage these previous results, we can improve the efficiency of constructing the output array.

It would be wise to find a way to take advantage of some **optimal substructure** by treating the results from integers 0 to $x - 1$ as potential subproblems of x . This is the beginning of a DP solution. Let $dp[x]$ represent the number of set bits in integer x .

A predictable way to access a subproblem of $dp[x]$ is to right-shift x by 1, effectively removing its LSB. This is the subproblem $dp[x \gg 1]$, and the only difference between this and $dp[x]$ is the LSB which was just removed.

As mentioned earlier, the LSB of x can be found using $x \& 1$. Therefore:

- If the LSB of x is 0, there is no difference in the number of bits between x and $x \gg 1$:



- If the LSB of x is 1, the difference in the number of bits between x and $x \gg 1$ is 1:



Therefore, we can obtain $dp[x]$ by using the result of $dp[x \gg 1]$ and adding the LSB to it:

$$dp[x] = dp[x \gg 1] + (x \& 1)$$

Now, we just need to know what our base case is.

Base

case

The simplest version of this problem is when n is 0. In this case, there are no set bits, so the number of set bits is 0. We can apply this base case by setting $dp[0]$ to 0.

After the base case is set, we populate the rest of the DP array by applying our formula from $dp[1]$ to $dp[n]$. The answer to the problem is then just the values in the DP array, containing the number of set bits for each number from 0 to n .

Try it yourself

Write your solution to the problem before checking the reference implementation.

Implementation - Dynamic Programming

Python

JavaScript

Java

```
import java.util.ArrayList;
```

```
public class Main {
    public ArrayList<Integer> hamming_weights_of_integers(int n) {
        // Base case: the number of set bits in 0 is just 0. We set dp[0] to 0 by
        // initializing the entire DP array to 0.
        int[] dp = new int[n + 1];
        for (int x = 1; x <= n; x++) {
            // 'dp[x]' is obtained using the result of 'dp[x >> 1]', plus the LSB of 'x'.
            dp[x] = dp[x >> 1] + (x & 1);
        }
        ArrayList<Integer> result = new ArrayList<>();
        for (int count : dp) {
            result.add(count);
        }
        return result;
    }
}
```

Complexity Analysis

Time complexity: The time complexity of `hamming_weights_of_integers_dp` is $O(n)O(n)$ since we populate each element of the DP array once.

Space complexity: The space complexity is $O(1)O(1)$ because no extra space is used, aside from the space taken up by the output, which is the DP array in this case.